



Data and Service Design

Designing for Data-Centric Services

Master of Product Service System Design

School of Design, Politecnico di Milano

Academic Year 2025-2026, 2nd semester

Nice to meet you



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Lecturer



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Course Organization

Abstract

The course intends to be a fundamental introduction to the notion of **data** for product-service system design students, providing them with **conceptual and critical tools**, as well as **practice-oriented** approaches and methods, to engage and work with data as a “**material**” in their design processes.

Course Framework

Theoretical Track

Narrowing down, then expanding



Start> End

Practical "Hands-on" Track

Each step builds on top of the previous



Theoretical Track

Narrowing down, then expanding



Start **End**

Practical "Hands-on" Track

Each step builds on top of the previous



Theoretical Track

Primarily based on lectures by professors and guest speakers, it first introduces interpretative lenses that use “data” as a way to describe the contemporary world. It then focuses on the relationship between data and services, and from there discusses broader governance and ethical implications.

Theoretical Track

Narrowing down, then expanding



Start> End

Practical "Hands-on" Track

Each step builds on top of the previous



Practical Track

Sequential and based on hands-on work in class: at each lesson, students will build upon what was done previously. This track is structured around 3 stages.

PT Stages

1. The basics of accessing and manipulating heterogeneous data sources, such as those available from the web and social media
2. Manipulating data into structured, analyzable formats and on developing appropriate data models to support exploration and analysis.
3. How to work with data from a representational perspective, and on the methods and tools that can be used to visualize complex data in ways that support exploration and decision-making.

Calendar

Calendar and class schedule

Class is held every week on Thursday
from 26th of February to 9th April
6 lessons in total

Class is suspended on the 26th of March

Class scheduling

- Morning: 9.30 - 13.00
- Afternoon: 14.30 - 18.00

Please, be on time!

	26 Feb	5 Mar	12 Mar	19 Mar	2 Apr	9 Apr
09.15 - 10.45	Lecture Product-service system design and data. Embarking on a Journey	Lecture (Counter)Narratives, Antagonistic Frames and Actions toward Datafication	Lecture Data-Driven Service vs. Data-Centric Services	Lecture Data-centric Organizations, Cultures and Governments	Lecture May Thou Data Cause Not Harm: Data Ethics principles	Class Assignment Presentations Schedule TBD
10.45 - 11.00	Morning Break					
11.00 - 12.45	Seminar Andrea Benedetti	Seminar Chiara Criscuolo	Seminar Davide Posillipo	Seminar Federico Bartolomucci	Seminar Petter Falk	
12.45 - 14.00	Lunch					
14.00 - 16.00	Practice Colab 101	Practice Datasets, APIs, and their aggregation	Practice Data preparation (+bit of statistics)	Practice Data Viz	Practice Recap and Group support	
16.00 - 16.15	Afternoon Break					
16.15 - 17.45	Practice Group Activity on Class Assignment	Practice Group Activity on Class Assignment	Practice Group Activity on Class Assignment	Practice Group Activity on Class Assignment	Practice Group Activity on Class Assignment	

Course Activities

Lectures

Theoretical lectures held by the professors on topics and concepts addressed by the course, followed by in class discussion.

Guest Lectures

Presentation by experts on
inspirational topics, specific thematics
and case studies on the intersection
between data and PSSD.

Guest Lectures

Andrea Benedetti

Università degli Studi di Milano

Chiara Criscuolo

Banca d'Italia

Davide Posillipo

Komebi Studio

Federico Bartolomucci

Politecnico di Milano

Petter Falk

Karlstad University

Classroom Practice

Guided **hands-on** on specific data-driven methods and tools.

For each **practice**, the class will work on a given practical example to be executed in class and linked to the **Final Assignment**.

Methods

- Python 101 (numpy, pandas, matplotlib) and basic use of APIs
- Dataset exploration and aggregation
- Data manipulation & preparation, including introductory statistical techniques
- Querying data and developing data visualizations

Tool

In-class work will be conducted using **Google Colab**.

You will use Colab to load and annotate datasets, clean and manipulate data, perform exploratory analysis, and develop visualizations.

Please make sure you can access this tool online through your Google Account.

Attendance Rules

Attendants
vs.
Non-Attendants

By the end of today, tell us whether you will be an **Attendant** or a **Non-Attendant**.

Be aware of the implications for the assignment and final evaluation.

Attendance Rules

Attendant students are required to attend **at least 4 out of the 6 classes**.

For this reason, you must sign the attendance register **twice a day**:
in the morning between **9:00 and 9:30**,
after the lunch break
between **14:00 and 14:30**.

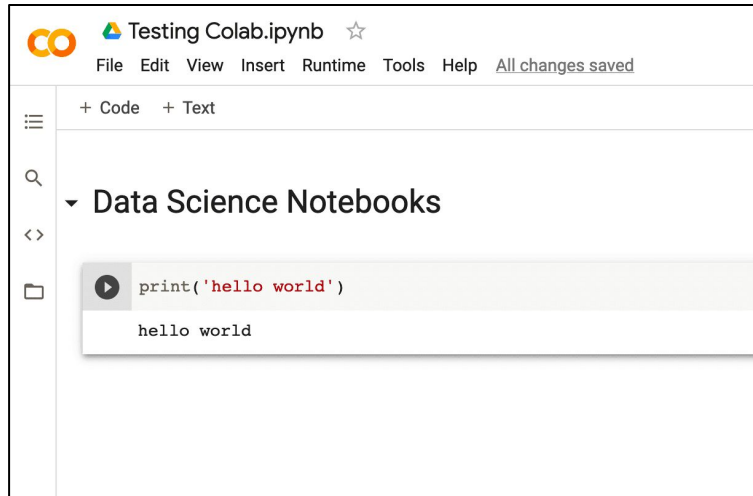
Final Assignment

Critical Data Notebook

The main deliverable of this course is a **Critical Data Notebook** – a reflective and analytical journal developed progressively throughout the course through Colab.

The objective of such a Notebook is to develop a documented and reflective output emerging from group data practices (analysis and visualization).

The mixed text-and-code format of a Colab Notebook allows students to critically document their reasoning, choices, and learning process alongside their code used to explore, analyze and visualize the data.



Critical Data Notebook

Students will work on broad **design opportunity areas** inspired by urban and territorial challenges contextualised in the areas of the **Municipality of Milan** and **Lombardy Region** (e.g., areas such as transportation, environment, and tourism).

Starting from the analysis of available data on these contexts and their services, students are expected to ideate **a basic service concept** — an original idea for a service — or a **re-design of an existing one**.

Critical Data Notebook Delivery

We will work on each module of the Final Assignment during the different lessons, so that the students will be guided in developing it by **working step-by-step in class**.

The final deliverable is due on **May 7th** (one month after the end of the course) and will be discussed individually by each student on the final exam day.

Critical Data Notebook Presentation

Attending students are required to present **their work-in-progress Assignment** to the faculty and the class on the last day of the course (April 9th).

Non-attending students are exempt from this requirement.

The presentation will be evaluated on the ability to communicate clearly and effectively to a mixed audience — conveying both the analytical process and the conceptual direction of the service being identified.

Evaluation

Evaluation

The final evaluation is individual for each student and is based on the combined assessment of:

1. the Class Assignment

- The quality of the submitted Critical Data Notebook
- The in-class presentation (Attending students only)

2. the Exam Colloquium.

Knowledge of the course topics and concepts
and ability to critically discuss them
with specific reference to the course bibliography

Evaluation

Each student will be asked to select 1 topic and 1 or 2 related readings from the Bibliography as starting topic to be discussed during the Exam Colloquium (additional topics may be asked by professors).

Course bibliography will be shared after this first lesson and updated after each lesson.

Evaluation weights

	1. Class Assignment		2. Exam Colloquium	
What is evaluated?	A. Quality of Assignment Delivered (Critical Data Notebook)	B. Capacity of presenting the in-progress Assignment as Group	A. Knowledge and critical discussion of topics/concepts with reference to the bibliography	B. Capacity of discussing theory in relation to Class Assignment
Attendants	Max 60%	Max 5%	Max 30%	Max 5%
Non-attendants	Max 50%	x	Max 40%	Max 10%

Course Tools

Syllabus

[Link](#)



PSSD PRODUCT-SERVICE
SYSTEM DESIGN



POLITECNICO | SCUOLA
MILANO 1863 | DEL DESIGN

Syllabus

Data and Service Design

Designing for Data-Centric Services

Faculty

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Website

dataservicedesign.github.io/syllabus

Data and Service Design

Editions | 2025-26 - 2024-25



Data and Service Design



Academic Year 2025-2026, 2nd semester

The course "*Data and Service Design: Designing Data-centric Services*" intends to be as a fundamental introduction to the notion of **data** for **product-service system design** students, providing them with conceptual and critical tools, as well as practice-oriented approaches and methods, to engage and work with data as a "material" in their design processes.

The understanding of the contemporary data landscape and its main topics will be developed through a double-track approach in which theoretical content and practical hands-on activities are carried out alongside each other.

Communications

messaggi.automatici

Official communications
(exams, rooms)

**WeBeep
Announcements and Forum**

Quick communication
(schedule changes, reminders)

Materials

WeBeep

Lesson materials
(slides, bibliography)

Delivery of Assignment
and Presentation

Group Formation

Access Miro at the link below
or via QR code

miro.com/app/board/uXjVG7tHERU=/?share link id=842761299774



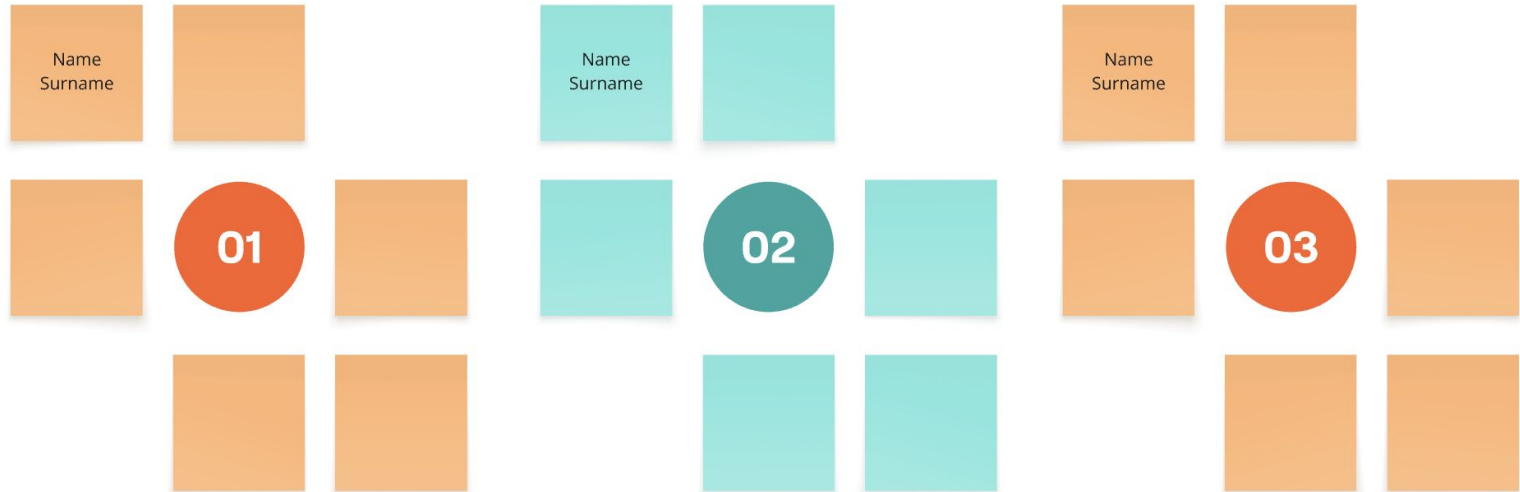
Add your name to the group

No more than 6 people per group!

Pick your team!

Take some time to talk with your classmates and form a team to support your Data Practices journey.
Write your name and surname on one of the sticky notes of the group you want to belong to.

DEADLINE: Monday March 2nd @ 10 PM - even better by today @ 2 PM



Now, please find the members of your group
and **sit together** in the classroom.
Each group is asked to choose one **spokesperson**.

Pick your team!

Take some time to talk with your classmates and form a team to support your Data Practices journey.
Write your name and surname on one of the sticky notes of the group you want to belong to.

DEADLINE: Monday March 2nd @ 10 PM - even better by today @ 2 PM

